PROBLEM 04

Plane at Constant Altitude

Original statement · Problems.pdf, p. 7

Coordinate Transformations

An aircraft moves at constant altitude y_0 with velocity v_0 in the x direction. At time $t = 0$ it is located at $x = 0$, and it begins to accelerate in the x direction at a constant rate a_x. Describe the position, velocity, and acceleration in polar coordinates r, θ, and answer the following questions:

(a) Show that: $\dot{r} = \frac{v(t)x(t)}{r(t)}$

(b) Show that: $\dot{\theta} = -\frac{v(t)y_0}{r^2(t)}$

(c) Show that: $\ddot{r} = \frac{v^2(t)y_0^2 + a_x x(t) r^2(t)}{r^3(t)}$

(d) Show that: $\ddot{\theta} = y_0 \frac{2v^2(t)x(t) - a_x r^2(t)}{r^4(t)}$

From these expressions form the acceleration vector $\mathbf{a} = a_r \mathbf{e}_r + a_\theta \mathbf{e}_\theta$. Repeat the problem using Cartesian coordinates.

Worked Solution

Constraints, Degrees of Freedom & Coordinates

- **System:** an aircraft moves at constant altitude y_0 with velocity v_0 along x. At $t = 0$ it is at $x = 0$ and begins to accelerate along x at a constant rate a_x. A fixed observer at the origin O describes its position with polar coordinates.

- **Hypotheses:** $y_0 > 0$, $v_0 \geq 0$ and a_x are constants.
- **Constraints and degrees of freedom:** the altitude is fixed, $y = y_0$ (holonomic, scleronomic, so $\dot{y} = \ddot{y} = 0$). The aircraft has $\mathrm{CDOF} = 2 - 1 = 1$, with the coordinate $x(t)$ given by the statement.
- **Coordinates:** Cartesian $x(t)$ for the motion, and polar (r, θ) about O for the description requested, with θ measured from $+\mathbf{i}$.

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{S}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ and $\mathbf{r} = x\,\mathbf{i} + y_0\,\mathbf{j}$.
- **Polar relations:** $r = \sqrt{x^2 + y_0^2}$ and $\tan\theta = y_0/x$; equivalently $\cos\theta = x/r$ and $\sin\theta = y_0/r$.
- **Polar basis.** $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis rotated by $\theta(t)$ about $\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} x/r & -y_0/r & 0 \\ y_0/r & x/r & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \frac{x^2 + y_0^2}{r^2} = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

The product has diagonal entries $(x^2 + y_0^2)/r^2 = 1$ (and 1) and off-diagonal entries $(xy_0 - y_0x)/r^2 = 0$.

- The inverse relation $\mathbf{i} = \cos\theta\,\mathbf{e}_r - \sin\theta\,\mathbf{e}_\theta$ is the one used in 3.4.

Derivation

3.1 Kinematics

Pedagogical Justification: the motion along x is uniformly accelerated, so $x(t)$ and $v(t)$ follow by integrating $\ddot{x} = a_x$ with the initial conditions $\dot{x}(0) = v_0$ and $x(0) = 0$ (Barrow's rule):

$$\dot{x}(t) = v_0 + \int_0^t a_x\,dt' = v_0 + [a_x t']_0^t = v_0 + a_x t - a_x \cdot 0 \equiv v(t), \qquad x(t) = 0 + \int_0^t (v_0 + a_x t')\,dt' = \left[v_0 t' + \frac{1}{2} a_x t'^2\right]_0^t = \left(v_0 t + \frac{1}{2} a_x t^2\right) - 0$$

- The polar position follows from the geometric relations of Phase 2: $\mathbf{r} = r\,\mathbf{e}_r$ with $r(t) = \sqrt{\left(v_0 t + \tfrac{1}{2}a_x t^2\right)^2 + y_0^2}$ and $\theta(t) = \operatorname{arccot}\left(\dfrac{v_0 t + \frac{1}{2}a_x t^2}{y_0}\right)$, with the branch $\operatorname{arccot} \in (0, \pi)$ (so $\theta(0) = \pi/2$, and θ stays in $(0, \pi)$ even if $a_x < 0$ makes x negative). The velocity and acceleration come from differentiating these relations.

(a) Proof that $\dot{r} = \dfrac{v(t)\,x(t)}{r(t)}$

> **Pedagogical Justification:** differentiate implicitly the relation $r^2 = x^2 + y_0^2$ with respect to time.

$$2r\dot{r} = 2x\dot{x} + 0 = 2x\,v(t) \implies \dot{r} = \frac{v(t)\,x(t)}{r(t)} \quad \blacksquare$$

(b) Proof that $\dot{\theta} = -\dfrac{v(t)\,y_0}{r(t)^2}$

> **Pedagogical Justification:** use $\cot\theta = x/y_0$, which is valid for every t (including $x = 0$, where $\tan\theta$ is infinite), and differentiate with the chain rule: $\frac{d}{dt}\cot\theta = -\csc^2\theta\,\dot{\theta}$ and $\csc^2\theta = 1/\sin^2\theta = r^2/y_0^2$.

$$-\csc^2\theta\,\dot{\theta} = \frac{\dot{x}}{y_0} = \frac{v(t)}{y_0} \implies -\frac{r^2}{y_0^2}\dot{\theta} = \frac{v}{y_0} \implies \dot{\theta} = -\frac{v(t)\,y_0}{r(t)^2} \quad \blacksquare$$

(c) Proof that $\ddot{r} = \dfrac{v^2(t)\,y_0^2 + a_x\,x(t)\,r^2(t)}{r(t)^3}$

> **Pedagogical Justification:** differentiate $\dot{r} = vx/r$ with the quotient rule $\left(\frac{u}{w}\right)' = \frac{\dot{u}w - u\dot{w}}{w^2}$, with $u = vx$ and $w = r$.

$$\dot{u} = \dot{v}x + v\dot{x} = a_x x + v^2, \qquad \dot{w} = \dot{r} = \frac{vx}{r}, \qquad \ddot{r} = \frac{\left(a_x x + v^2\right) r - vx\,\frac{vx}{r}}{r^2} = \frac{\left(a_x x + v^2\right) r^2 - v^2 x^2}{r^3}$$

$$\left(a_x x + v^2\right) r^2 - v^2 x^2 = a_x x r^2 + v^2\left(r^2 - x^2\right) = a_x x r^2 + v^2 y_0^2 \implies \ddot{r} = \frac{v^2 y_0^2 + a_x\, x\, r^2}{r^3}$$ ■

(d) Proof that $\ddot{\theta} = y_0 \dfrac{2v^2(t)\, x(t) - a_x\, r^2(t)}{r(t)^4}$

> **Pedagogical Justification:** differentiate $\dot{\theta} = -y_0\, v\, r^{-2}$ with the quotient rule, using $\dot{v} = a_x$ and $\frac{d}{dt}\left(r^2\right) = 2r\dot{r} = 2vx$:

$$\ddot{\theta} = -y_0\, \frac{\dot{v} r^2 - v\, \frac{d}{dt}(r^2)}{r^4} = -y_0\, \frac{a_x r^2 - 2v^2 x}{r^4} = y_0\, \frac{2v^2 x - a_x r^2}{r^4}$$ ■

Position, velocity and acceleration in polar coordinates

- **Position and velocity:** $\mathbf{r} = r\, \mathbf{e}_r$ and, with (a) and (b),

$$\mathbf{v} = \dot{r}\, \mathbf{e}_r + r\dot{\theta}\, \mathbf{e}_\theta = v\left(\frac{x}{r}\, \mathbf{e}_r - \frac{y_0}{r}\, \mathbf{e}_\theta\right)$$

- **Acceleration** $\mathbf{a} = a_r \mathbf{e}_r + a_\theta \mathbf{e}_\theta$ from the polar formula:

$$a_r = \ddot{r} - r\dot{\theta}^2 = \frac{v^2 y_0^2 + a_x x r^2}{r^3} - r\left(\frac{v y_0}{r^2}\right)^2 = \frac{v^2 y_0^2 + a_x x r^2}{r^3} - \frac{v^2 y_0^2}{r^3} = \frac{a_x x r^2}{r^3} = a_x \frac{x}{r}$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = \frac{y_0\left(2v^2 x - a_x r^2\right)}{r^3} + 2\,\frac{vx}{r}\left(-\frac{v y_0}{r^2}\right) = \frac{y_0\left(2v^2 x - a_x r^2\right)}{r^3} - \frac{2v^2 x y_0}{r^3} = -a_x \frac{y_0}{r} \implies \mathbf{a} = a_x\left(\frac{x}{r}\, \mathbf{e}_r - \frac{y_0}{r}\, \mathbf{e}_\theta\right)$$

3.2 Forces

Not applicable: kinematics-only problem. The motion law along x is given.

3.3 Equations of Motion

Not applicable: kinematics-only problem. The acceleration a_x is prescribed.

3.4 Integration and Answers

Repeating the problem in Cartesian coordinates

> **Pedagogical Justification:** in the fixed Cartesian basis the unit vectors are constant, so the description is immediate; the polar result must agree with it through the change of basis.

$$\mathbf{r} = x(t)\,\mathbf{i} + y_0\,\mathbf{j}, \qquad \mathbf{v} = v(t)\,\mathbf{i}, \qquad \mathbf{a} = a_x\,\mathbf{i}$$

- Converting with $\mathbf{i} = \cos\theta\,\mathbf{e}_r - \sin\theta\,\mathbf{e}_\theta = \frac{x}{r}\,\mathbf{e}_r - \frac{y_0}{r}\,\mathbf{e}_\theta$:

$$\mathbf{v} = v\left(\frac{x}{r}\,\mathbf{e}_r - \frac{y_0}{r}\,\mathbf{e}_\theta\right), \qquad \mathbf{a} = a_x\left(\frac{x}{r}\,\mathbf{e}_r - \frac{y_0}{r}\,\mathbf{e}_\theta\right)$$

These equal the polar results of 3.1 identically.

Interpretation & Checks

- **Dimensional check:** $[\dot r] = [vx/r] = \mathrm{m/s}$ and $[\dot\theta] = [vy_0/r^2] = \mathrm{s^{-1}}$; $[\ddot r] = \dfrac{(\mathrm{m/s})^2\mathrm{m}^2}{\mathrm{m}^3} = \mathrm{m/s^2}$ and $[\ddot\theta] = \dfrac{\mathrm{m}\cdot(\mathrm{m/s})^2\cdot\mathrm{m}}{\mathrm{m}^4} = \mathrm{s^{-2}}$.
- **Limit $x \to \infty$ (aircraft far away):** $r \to x$, so $\dot r \to v$, $\dot\theta \to 0$, $a_r \to a_x$ and $a_\theta \to 0$: the line of sight aligns with the flight direction and the motion is purely radial.
- **Aircraft overhead ($x = 0, t = 0$):** $\dot r = 0, \dot\theta = -v_0/y_0, a_r = 0$ and $a_\theta = -a_x$; with $\theta = \pi/2$, $\mathbf{a} = a_x\mathbf{i} = -a_x\mathbf{e}_\theta$.
- **Invariance of the modulus:** $a_r^2 + a_\theta^2 = a_x^2\,\dfrac{x^2 + y_0^2}{r^2} = a_x^2$, equal to $\|a_x\mathbf{i}\|^2$ because $[{}_0R_1]$ is orthogonal.

- **Check against the official key:** `Problems.pdf` has no entry for Problem 4 in its solution keys, but the statement gives the four formulas (a) to (d) as targets, and they are proved above. Sympy confirms the four identities and $a_r = a_x x/r$, $a_\theta = -a_x y_0/r$ with zero residual.

Source: Problems.pdf